

Homework 1

Grant Talbert

January 30, 2026

Boston University

Problem 1. Let $a \in \mathbb{R}$ be irrational, and consider the set

$$G := \left\{ \begin{pmatrix} e^{it} & 0 \\ 0 & e^{ita} \end{pmatrix} \mid t \in \mathbb{R} \right\}.$$

Show that the closure of G in $\mathrm{GL}(2, \mathbb{C})$ is the torus

$$T := \left\{ \begin{pmatrix} e^{i\theta} & 0 \\ 0 & e^{i\phi} \end{pmatrix} \mid \theta, \phi \in \mathbb{R} \right\}.$$

Proof. First we show that $T \subseteq \overline{G}$. Let

$$A = \begin{pmatrix} e^{i\theta} & 0 \\ 0 & e^{i\phi} \end{pmatrix} \in T,$$

and let $U \subseteq \mathrm{GL}(2, \mathbb{C})$ be an open neighborhood of A . Since the product topology has a basis given by products of open sets in $\mathbb{C}$, which has a basis given by open balls, we can take U to be a product $V_1 \times V_2 \times V_3 \times V_4$ for some open sets $V_i \subseteq \mathbb{C}$, $1 \leq i \leq 4$ with $e^{i\theta} \in V_1$, $e^{i\phi} \in V_2$, and $0 \in V_3, V_4$. It suffices to show $U \cap G \neq \emptyset$. For simplicity of notation, we identify matrices with tuples via the identification

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \iff (a, b, c, d).$$

It is useful to recall [Hal15] 1.6 exercise 9, which states that $\{e^{i2\pi na} \mid n \in \mathbb{Z}\}$ is dense in S^1 whenever a is irrational. Define $t_n := \theta + 2\pi n$ for each $n \in \mathbb{Z}$, and note that

$$e^{it_n} = e^{i\theta + i2\pi n} = e^{i\theta} e^{i2\pi n} = e^{i\theta}.$$

We claim there exists n such that $(e^{it_n}, e^{it_n a}, 0, 0) \in U$. Since $e^{it_n} = e^{i\theta} \in V_1$, it suffices to show that there exists an n such that $e^{it_n a} \in V_2$. Note that

$$e^{it_n a} = e^{i(\theta + 2\pi n)a} = e^{i\theta a} e^{i2\pi na}.$$

Note that multiplication by $e^{i\theta a}$ is a homeomorphism. Hence there is $e^{it_n a} \in V_2$ if and only if there is $e^{i2\pi na} \in e^{-i\theta a} V_2$. Since $e^{-i\theta a} V_2$ is the preimage of the open set V_2 under the continuous map $p \mapsto e^{i\theta a} p$, it is open. By density of $\{e^{i2\pi na} \mid n \in \mathbb{Z}\}$ in S^1 , since $e^{-i\theta a} V_2 \cap S^1$ is open in S^1 , there exists $e^{i2\pi na}$ in $e^{-i\theta a} V_2 \cap S^1 \subseteq e^{-i\theta a} V_2$. Thus $(e^{i\theta}, e^{i\phi}, 0, 0) \in \overline{G}$.

Now we show $\overline{G} \subseteq T$. First note that by definition, $G \subseteq T$. Next note that if $A \in \overline{G}$, then every open neighborhood of A intersects G , and thus T . Hence $A \in \overline{T}$. We complete the proof by showing T is closed. Note that S^1 is closed and bounded when viewed in $\mathbb{R}^2 \cong \mathbb{C}$, and by Heine-Borel's theorem, S^1 is compact. By Tychonoff's theorem, the product $S^1 \times S^1$ is compact in $\mathbb{C}^2$. The inclusion map $\mathbb{C}^2 \hookrightarrow \mathbb{C}^4$ defined by $(w, z) \mapsto (w, z, 0, 0)$ is clearly continuous, and the image of $S^1 \times S^1$ under this inclusion is T . Since images of continuous maps preserve compactness, T is compact in $\mathbb{C}^4$. Since $\mathbb{C}^4$ is Hausdorff, and compact sets are closed in Hausdorff spaces, T is closed in $\mathbb{C}^4$. Since passing to the subspace topology preserves closedness, T is closed in $\mathrm{GL}(2, \mathbb{C})$. Thus $T = \overline{T}$, and since $\overline{G} \subseteq \overline{T}$, we have $\overline{G} \subseteq T$. ■

Problem 2. *Prove that the Cayley transformation*

$$A \mapsto (I - A)(I + A)^{-1}$$

sends orthogonal matrices in $\mathrm{SO}(2)$ with eigenvalue $\neq -1$ to skew-symmetric matrices in $\mathfrak{so}(2)$, and vice versa.

Proof. Let $A \in \mathrm{SO}(2)$ without eigenvalue -1 . Since λ is an eigenvalue if and only if the matrix $A - \lambda I$ is not invertible, it follows that $A - (-1)I = A + I$ must be invertible. Define $Q := (I - A)(I + A)^{-1}$. We must show $Q^t = -Q$. Since A is orthogonal, $A^t = A^{-1}$. By the various properties of transpose,

$$\begin{aligned} Q^t &= ((I - A)(I + A)^{-1})^t \\ &= ((I + A)^{-1})^t (I - A)^t \\ &= (I^t + A^t)^{-1} (I^t - A^t) \\ &= (I + A^{-1})^{-1} (I - A^{-1}) \\ &= ((A + I)A^{-1})^{-1} (A - I)A^{-1} \\ &= A(A + I)^{-1}(A - I)A^{-1} \\ &\stackrel{!}{=} (A + I)^{-1}AA^{-1}(A - I) \\ &= (A + I)^{-1}(A - I) \\ &\stackrel{!!}{=} (A - I)(A + I)^{-1} \\ &= -(I - A)(I + A)^{-1} \\ &= -Q. \end{aligned}$$

The equalities marked with ! and !! require some minor justification. For the first, note that since A commutes with I and itself, it thus commutes with $A + I$ and $A - I$. Note that if $BC = CB$, then $C^{-1}B = C^{-1}BCC^{-1} = C^{-1}CBC^{-1} = BC^{-1}$. Applying this here yields $A(A + I)^{-1} = (A + I)^{-1}A$ and $(A - I)A^{-1} = A^{-1}(A - I)$. The equality marked with !! follows by this same logic: since A, I commute, the linear combinations $A - I$ and $A + I$ commute, and thus so do their inverses.

Now let $A \in \mathfrak{so}(2)$, meaning $A^t = -A$. Skew symmetric matrices have purely imaginary eigenvalues, so by the same logic as before $I + A$ is invertible. Define $Q := (I - A)(I + A)^{-1}$. We must show $Q^tQ = I$ and $\det Q = 1$.

$$\begin{aligned} Q^T &= ((I - A)(I + A)^{-1})^t \\ &= ((I + A)^{-1})^t (I - A)^t \\ &= (I^t + A^t)^{-1} (I^t - A^t) \\ &= (I - A)^{-1} (I + A) \\ &= ((I + A)^{-1} (I - A))^{-1} \\ &= ((I - A)(I + A)^{-1})^{-1} \\ &= Q^{-1} \end{aligned}$$

Commuting $I - A$ and $(I + A)^{-1}$ follows by the same logic as before. Thus Q is indeed orthogonal.

Now using the fact that determinant preserves multiplication, and that $\det B^t = \det B$,

$$\begin{aligned}
\det Q &= \det(I - A) \det((I + A)^{-1}) \\
&= \det(I - A) (\det(I + A))^{-1} \\
&= \frac{\det(I - A)}{\det(I + A)} \\
&= \frac{\det(I - A)^t}{\det(I + A)} \\
&= \frac{\det(I^t - A^t)}{\det(I + A)} \\
&= \frac{\det(I + A)}{\det(I + A)} \\
&= 1.
\end{aligned}$$

Finally, we show that Q does not have -1 as an eigenvalue. Suppose for contradiction that there exists $v \neq 0$ with $Qv = -v$. Then

$$\begin{aligned}
&(I - A)(I + A)^{-1}v = -v \\
\implies (I + A)^{-1}(I - A)v &= -v \\
\implies (I - A)v &= -(I + A)v \\
\implies v - Av &= -v - Av \\
\implies v &= -v \\
\implies v &= 0.
\end{aligned}$$

This contradicts $v \neq 0$, meaning -1 is not an eigenvalue of Q . ■

References

[Hal15] Brian Hall. *Lie Groups, Lie Algebras, and Representations*. Springer GTM, 2015.